

Launch Amazon EC2 in a Non-Default VPC

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Purpose

In this lab, I created a VPC without default settings. In other words we adjusted the settings when creating a VPC to configure the VPC to a company's specific needs.

Lab Walkthrough (with Documentation)

Step 1: Create a VPC

Create VPC Info

A VPC is an isolated portion of the AWS cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block Info

IPv6 CIDR block Info

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy Info

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	VRTest VPC	Remove

You can add 49 more tags.

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Step 2: Create a Public Subnet

Create subnet [Info](#)

VPC

VPC ID
Create subnets in this VPC.

vpc-0d2a3e32c60d8bc19 (VCTest VPC) ▼

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

Public Subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US East (N. Virginia) / us-east-1a ▼

IPv4 CIDR block [Info](#)

Q 10.0.1.0/24 X

Step 3: Create a Private Subnet

Create subnet [Info](#)

VPC

VPC ID
Create subnets in this VPC.

vpc-0d2a3e32c60d8bc19 (VCTest VPC) ▼

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

Private Subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

US East (N. Virginia) / us-east-1a ▼

IPv4 CIDR block [Info](#)

Q 10.0.2.0/24 X

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Step 4: Create an Internet Gateway

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Remove

Add new tag

You can add 49 more tags.

CancelCreate internet gateway

Step 5: Attach the Internet Gateway to your VPC

Attach to VPC (igw-0320bd80e43f88d57) [Info](#)

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

AWS Command Line Interface command

CancelAttach internet gateway

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Step 6: Create a Public Route Table

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

vpc-0d2a3e32c60d8bc19 (VRTest VPC) ▼

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

X

Value - optional

X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Step 7: Create a Private Route Table

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

vpc-0d2a3e32c60d8bc19 (VRTest VPC) ▼

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

X

Value - optional

X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

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Step 8: Add a Route to the Public Route Table

Edit routes

Edit routes

Destination

10.0.0.0/16

Target

Q local

X

Status

Active

Propagated

No

Edit routes

Destination

Q 0.0.0.0/0

X

Target

Q igw-0320bd80e43f88d57

X

Status

-

Propagated

No

Remove

Add route

Cancel

Preview

Save changes

Step 9: Associate the Public Route Table to the Public Subnet

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Q Filter subnet associations

< 1 > ⚙

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public Subnet	subnet-04bd91713693e97bc	10.0.1.0/24	-	Main (rtb-026efcfad95a9b26)
☐	Private Subnet	subnet-04b6e5229dcb59acd	10.0.2.0/24	-	Main (rtb-026efcfad95a9b26)

Selected subnets

subnet-04bd91713693e97bc / Public Subnet

X

Cancel

Save associations

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Step 10: Add a Route to the Private Route Table

Edit routes

Edit routes

Destination

10.0.0.0/16

Target

Q local

X

Status

Active

Propagated

No

Edit routes

Destination

Q 0.0.0.0/0

X

Target

Q igw-0320bd80e43f88d57

X

Status

-

Propagated

No

Remove

Add route

Cancel

Preview

Save changes

Step 11: Associate the Private Route Table to the Private Subnet

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Q Filter subnet associations

< 1 > ⌕

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	Public Subnet	subnet-04bd91713693e97bc	10.0.1.0/24	-	rtb-0550eca6182a4b585 / Public Route Table
☒	Private Subnet	subnet-04b6e5229dcb59acd	10.0.2.0/24	-	Main (rtb-026efcfad95a9b26)

Selected subnets

subnet-04b6e5229dcb59acd / Private Subnet X

Cancel

Save associations

Step 12: Create an EC2 Instance

Step 1: Choose an Amazon Machine Image (AMI)

[Cancel and Exit](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Q Search for an AMI by entering a search term e.g. "Windows"

Search by Systems Manager parameter

Quick Start

My AMIs

AWS Marketplace

Community AMIs

☐ Free tier only ⓘ

**Amazon Linux**
Free tier eligible

Amazon Linux 2 AMI (HVM), SSD Volume Type - ami-048f6ed62451373d9 (64-bit x86) / ami-00315de4391ce4f6d (64-bit Arm)
Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is approaching end of life on December 31, 2020 and has been removed from this wizard.
Root device type: ebs Virtualization type: hvm ENA Enabled: Yes

Select

☒ 64-bit (x86)
☐ 64-bit (Arm)

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Step 13: Choose an Instance Type

Step 2: Choose an Instance Type

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instances are virtual servers that can run applications. They have varying combinations of CPU, memory, storage, and networking capacity, and give you the flexibility to choose the appropriate mix of resources for your applications. [Learn more](#) about instance types and how they can meet your computing needs.

Filter by: All instance families Current generation [Show/Hide Columns](#)

Currently selected: t2.micro (- ECUs, 1 vCPUs, 2.5 GHz, -, 1 GiB memory, EBS only)

	Family	Type	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance	IPv6 Support
☐	t2	t2.nano	1	0.5	EBS only	-	Low to Moderate	Yes
☒	t2	t2.micro Free tier eligible	1	1	EBS only	-	Low to Moderate	Yes

Step 14: Configure Instance Details

Step 3: Configure Instance Details

Configure the instance to suit your requirements. You can launch multiple instances from the same AMI, request Spot instances to take advantage of the lower pricing, assign an access management role to the instance, and more.

Number of instances i [Launch into Auto Scaling Group](#) i

Purchasing option i ☐ Request Spot instances

Network i vpc-0d2a3e32c60d8bc19 | VRTest VPC [Create new VPC](#)

Subnet i subnet-04bd91713693e97bc | Public Subnet | us-east-1 [Create new subnet](#)
251 IP Addresses available

Auto-assign Public IP i Enable

▼ Advanced Details

Enclave i ☐ Enable

Metadata accessible i Enabled

Metadata version i V1 and V2 (token optional)

Metadata token response hop limit i 1

User data i ☒ As text ☐ As file ☐ Input is already base64 encoded

```
#!/bin/bash
yum -y install httpd
systemctl enable httpd
systemctl start httpd
echo "<html><h1>Hello Earthling, Take me to your leader! </h1></html>"
>/var/www/html/index.html
```

[Cancel](#) [Previous](#) [Review and Launch](#) [Next: Add Storage](#)

Step 15: Add Two Tags

Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver.

A copy of a tag can be applied to volumes, instances or both.

Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

Key	Value	Instances	Volumes	Network Interfaces	
Name	VR Testing Server	☒	☒	☒	☒
Department	Development	☒	☒	☒	☒

[Add another tag](#) (Up to 50 tags maximum)

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Step 16: Configure a Security Group

Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group: ☒ Create a new security group
☐ Select an existing security group

Security group name:

Description:

Type ⓘ	Protocol ⓘ	Port Range ⓘ	Source ⓘ	Description ⓘ
SSH ▾	TCP	22	Custom ▾ 0.0.0.0/0	e.g. SSH for Admin Desktop ✕
HTTP ▾	TCP	80	Anywhere ▾ 0.0.0.0/0, ::/0	e.g. SSH for Admin Desktop ✕

Step 17: Proceed Without a Key Pair

Select an existing key pair or create a new key pair ✕

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

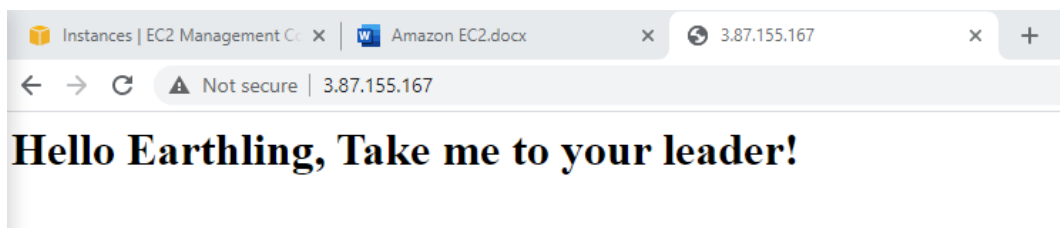
Proceed without a key pair ▾

☒ I acknowledge that I will not be able to connect to this instance unless I already know the password built into this AMI.

[Cancel](#)

[Launch Instances](#)

Step 18: Check If Your Website Works



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Problems

I had trouble accessing the webpage using the IP address. I looked through my configurations and code again and everything seemed right. I checked again with a peer, and my configurations appeared to match their configurations as well. However, after letting some time pass as well as using a different device that was not my school issued laptop, I was able to access the website.

Lab Questions

1. **What is your default VPC? Why did you use it?** My default VPC is VRTest VPC and I would use it to immediately start launching Amazon EC2 instances into your default VPC.
2. **When creating your public subnet, what must you do to make the subnet public?** To make a subnet public when creating it, you must select Enable auto-assign public IPv4.
3. **What role does the route table play in a VPC?** A routing table's role in a VPC is to contain the routes that will determine where the traffic will go.
4. **If you do not specify the subnet and availability zone when creating an Amazon EC2 instance, what happens?** If you do not specify the subnet and availability zone when creating an Amazon EC2 instance, then the availability zone will be set to "no preference" and the subnet will be created in an availability zone that is available.
5. **What is the 'auto-assign public IP' default setting?** The default setting for the auto-assign public IP is Disable.
6. **What will happen if you do not adjust the auto-assign public IP setting to "enable"?** If you don't adjust the auto-assign public IP setting to enable, then it won't assign the VPC a public IP address.
7. **Describe what a security group is and why it's important.** A security group is basically a virtual firewall for your EC2 instances that controls incoming and outgoing traffic. They are important because they protect your applications and data.
8. **Why did you create a private subnet? When would you use this?** I created a private subnet because it connects the VPC to the internet and other AWS services. I would use this to prepare for if one of my instances were hacked as it would be more difficult to hack into instances in the private subnet.
9. **What is the purpose of the internet gateway? What's the result if you don't create and attach an internet gateway to your VPC?** The purpose of the internet gateway is to provide a target in your VPC route tables for internet-routable traffic and to perform NAT for instances that have been assigned public IPv4 addresses. If you don't create and attach an internet gateway to your VPC then there will be no target destination for the VPC route tables and your VPC will not be able to access the internet.

Bonus Activity #1

You can create Key Pairs through the “Key Pairs” section in EC2

EC2 > Key pairs > Create key pair

Create key pair

Key pair

A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.

Name

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

File format

☐ pem
For use with OpenSSH

☒ ppk
For use with PuTTY

Tags (Optional)

No tags associated with the resource.

You can add 50 more tags.

Cancel

Bonus Activity #2

Step 1: Terminate Your Amazon EC2 Instance

Terminate instance?

×

Instance IDs

 i-0bb460c379957ba5e (VR Testing Server)

To confirm that you want to terminate the instance, choose the *terminate* button below.
Terminating the instance cannot be undone

Cancel

Terminate

Step 2: Delete Your Amazon VPC

Delete VPC

×

 VRTest VPC

 vpc-0d2a3e32c60d8bc19

✔ Available

✔ Will also be deleted

The following 6 resources will also be deleted permanently and cannot be recovered later:

Name	Resource ID	State
-	sg-03ca35aef70bf8aa2	-
Private Route Table	rtb-0925feef7e925715c	-
Private Subnet	subnet-04b6e5229dcb59acd	✔ Available
Public Route Table	rtb-0550eca6182a4b585	-
Public Subnet	subnet-04bd91713693e97bc	✔ Available
VRTest IGW	igw-0320bd80e43f88d57	✔ Available

To confirm deletion, type *delete* in the field:

delete

Cancel

Delete

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